A Hardy space approximation supporting zero-free half-planes for the ζ-function

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Abstract

By a theorem of Bagchi, the Riemann Hypothesis (RH) holds if and only if $E(s) = 1/s$ belongs to the closed linear span of $G_k(s) = (k^{-s} - k^{-1})\zeta(s)/s$, $k \geq 2$, in the Hardy space $H^2(\mathbb{C}_{1/2})$, where $\mathbb{C}_\alpha$ denotes the half-plane $\mathrm{Re}(s) > \alpha$. The formal Möbius expansion $E = \sum_{k\geq 2} \mu(k) G_k$ corresponds to a series that Báez-Duarte proved to diverge in L^2, so its partial sums S_n do not converge on the critical line. We study this expansion in the scale of spaces $H^2(\mathbb{C}_\alpha)$, where its convergence forces ζ to be zero-free in $\mathbb{C}_\alpha$. We prove that $S_n \to E$ in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$, by transferring a convergence result of Noor from the Hardy space of the disk to the half-planes, and that a single zero of ζ in $\mathbb{C}_\alpha$ keeps the error $d_n(\alpha)^2$ of best approximation of E by $G_2, \ldots, G_n$ uniformly away from zero. For $1/2 \leq \alpha \leq 1$ we express the Möbius error $s_n(\alpha) = \|S_n - E\|_\alpha^2$, where $\|\cdot\|_\alpha$ is the norm of $H^2(\mathbb{C}_\alpha)$, as a series with elementary terms, free of ζ, and compute it for $n \leq 10^6$: $s_n(1/2)$ oscillates without decaying, whereas for $\alpha > 1/2$ the oscillations decay at a rate close to $n^{1-2\alpha}$, much faster than the bound $O(n^{-(\alpha-1/2)/3})$ that follows under RH from estimates of Balazard and Saias. On the critical line, $d_n(1/2)^2$ behaves like $0.046/\log n$ for $100 \leq n \leq 5120$, in agreement with earlier computations by Báez-Duarte, Balazard, Landreau and Saias; under RH, the obstruction there thus lies in the Möbius coefficients rather than in the approximating subspaces.

Keywords: Riemann hypothesis, Nyman–Beurling criterion, Báez-Duarte criterion, Möbius function, zero-free half-planes, Hardy spaces, numerical approximation.

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1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$. Its implications for the distribution of prime numbers make it one of the central open problems in mathematics.

A functional-analytic approach to RH was initiated by Nyman and Beurling [9, 14]. Writing $\{y\}$ for the fractional part of y and $\chi = \chi_{(0,1]}$, their criterion states that RH holds if and only if χ can be approximated in $L^2(0,\infty)$ by linear combinations of the dilations $x \mapsto \{\theta/x\}$, $0 < \theta < 1$;

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see [3, Theorem 1.1 and Remark 1.1] for this form. Báez-Duarte [2, 3] examined the most natural candidate for such an approximation. Möbius inversion of the identity $\lfloor 1/x \rfloor = \sum_{j\geq 1} \chi(jx)$ [3, (2.4)] gives $\chi(x) = \sum_{j\geq 1} \mu(j)\lfloor 1/(jx)\rfloor$ [3, (2.6)], where μ is the Möbius function, and writing $\lfloor y \rfloor = y - \{y\}$ and using $\sum_j \mu(j)/j = 0$ one arrives at

$$\chi(x) = -\sum_{j=1}^{\infty} \mu(j)\Big\{\frac{1}{jx}\Big\}. \tag{1.1}$$

This series converges pointwise and in $L^1(0,1)$ [3, §2.2], but Báez-Duarte proved that it does not converge in $L^2(0,\infty)$ [3, Theorem 2.2]: the Möbius coefficients, natural as they are, fail precisely in the norm in which the criterion is formulated.

The same coefficients run through the subsequent literature. Báez-Duarte later strengthened the Nyman–Beurling criterion by replacing the uncountable family of dilations with the countable family $x \mapsto \{1/(kx)\} - \{1/x\}/k$, $k \geq 2$ [4]. In an equivalent weighted ℓ^2-formulation, in which the constant sequence $\mathbf{1}$ is approximated by the remainder sequences $r_k(j) = j \bmod k$, Balazard and Saias [7] asked whether the coefficients $c_{n,k}$ of the orthogonal projection of $\mathbf{1}$ onto $\operatorname{span}\{r_k : 2 \leq k \leq n\}$ satisfy $\lim_{n\to\infty} c_{n,k} = -\mu(k)/k$ for every $k \geq 2$, and Weingartner [20] proved that this is equivalent to RH. In the Hardy space $H^2(\mathbb{D})$ of the unit disk, Noor [13] showed that RH holds if and only if the constant function $\mathbf{1}$ belongs to the closed linear span of

$$h_k(z) = \frac{1}{1-z}\log\left(\frac{1 + z + \cdots + z^{k-1}}{k}\right), \quad k \geq 2,$$

and proved that $\sum_{k\geq 2} \frac{\mu(k)}{k}(I - S)h_k$ converges to $1 - z$, where S is the shift operator. Ghosh, Kremnizer, Noor and Santos [11] developed a general approach in which the density of $\operatorname{span}\{h_k\}$ in suitable spaces of analytic functions yields zero-free half-planes, and showed in particular that a Möbius series in the h_k converges in $H^q(\mathbb{D})$, $0 < q < 1$, which gives the half-plane $\operatorname{Re}(s) > 1$; see also [11, pp. 5–6] for an overview of the many equivalent reformulations of the Nyman–Beurling and Báez-Duarte criteria. Throughout, the Möbius coefficients are the canonical candidates, and what is at stake is the norm in which convergence is measured.

In this paper we pursue this question in the Hardy spaces of half-planes. For $\alpha \in \mathbb{R}$ let $\mathbb{C}_\alpha = \{s \in \mathbb{C} : \operatorname{Re}(s) > \alpha\}$. Bagchi [6] showed that RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of

$$G_k(s) = \big(k^{-s} - k^{-1}\big)\frac{\zeta(s)}{s}, \quad k \geq 2,$$

in $H^2(\mathbb{C}_{1/2})$. The Mellin transform carries χ to E and $\{1/(kx)\} - \{1/x\}/k$ to $-G_k$ (see Subsection 4.1), and under this correspondence (1.1) becomes the formal identity

$$E = \sum_{k=2}^{\infty} \mu(k)\, G_k, \tag{1.2}$$

which also follows formally from $\sum_k \mu(k)k^{-s} = 1/\zeta(s)$ and $\sum_k \mu(k)/k = 0$. By Plancherel's theorem the norm of $H^2(\mathbb{C}_{1/2})$ corresponds to the norm of $L^2(0,1)$, and Báez-Duarte's theorem

shows that the partial sums $S_n = \sum_{k=2}^{n} \mu(k) G_k$ do not converge to E in $H^2(\mathbb{C}_{1/2})$ (Remark 4.2). The space $H^2(\mathbb{C}_{1/2})$, however, is only the first of the scale $H^2(\mathbb{C}_\alpha)$, and since $\|F\|_\alpha \le \|F\|_{1/2}$ for $\alpha \ge 1/2$, where $\|\cdot\|_\alpha$ denotes the norm of $H^2(\mathbb{C}_\alpha)$, convergence becomes easier as α increases. This raises the question that motivates the present work: for which α does (1.2) converge in $H^2(\mathbb{C}_\alpha)$, and what does its convergence tell us about ζ?

The starting point is the following simple observation, which ties convergence in a half-plane to the zeros of ζ in that half-plane (Proposition 3.4): for every $\alpha \ge 1/2$,

$$E \in \overline{\operatorname{span}}\{G_k : k \ge 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \ne 0, \quad \forall\, s \in \mathbb{C}_\alpha.$$

In particular, the convergence of the Möbius series (1.2) in $H^2(\mathbb{C}_\alpha)$ is a sufficient condition for the half-plane $\mathbb{C}_\alpha$ to be free of zeros, and Báez-Duarte's obstruction is the case $\alpha = 1/2$ of a whole scale of such conditions. We then prove (Theorem 3.2) that $S_n \to E$ in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$; in the scale $H^2(\mathbb{C}_\alpha)$ this is the counterpart of the result of [11] for $H^q(\mathbb{D})$. The proof extends Noor's convergence result to a family of functions $g_{k,\tau}$ in $H^2(\mathbb{D})$ and transfers it, through the Hilbert space $\mathcal{H}^2$ of Dirichlet series, to $H^2(\mathbb{C}_\alpha)$ by a bounded operator $\mathcal{M}^\tau : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2+\tau})$. A quantitative counterpart also holds. For $n \ge 2$ and $\alpha > 0$ let

$$d_n(\alpha)^2 := \inf_{c_2,\dots,c_n \in \mathbb{C}} \Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2$$

denote the optimal approximation error. Theorem 3.5 shows that a single zero of ζ in $\mathbb{C}_\alpha$ keeps $d_n(\alpha)^2$ bounded away from 0, uniformly in n, and Corollary 3.6 shows that the threshold $\alpha_c = \inf\{\alpha \ge 1/2 : d_n(\alpha)^2 \to 0\}$ satisfies $1/2 \le \Theta \le \alpha_c \le 1$, where Θ is the supremum of the real parts of the non-trivial zeros of ζ, with $\alpha_c = 1/2$ if and only if RH holds.

Between the critical line, where (1.2) diverges, and the half-plane $\operatorname{Re}(s) > 1$, where it converges, lies the critical strip, and it is there that the question becomes decisive. By Proposition 3.4, a proof of convergence for some $\alpha < 1$ would yield a zero-free half-plane far beyond the known zero-free regions of ζ, so we explore the strip numerically. An exact representation of the Möbius error

$$s_n(\alpha) = \|S_n - E\|_\alpha^2$$

as a series with elementary terms (Proposition 4.1), which avoids both the zeta function and numerical quadrature, allows us to compute $s_n(\alpha)$ up to $n = 10^6$ for $1/2 \le \alpha \le 1$. On the critical line $s_n(1/2)$ oscillates without decaying, consistently with Báez-Duarte's theorem, which implies that $s_n(1/2) \not\to 0$ (Remark 4.2). For every $\alpha > 1/2$, on the other hand, our computations show that the oscillations decay, at a rate close to $n^{1-2\alpha}$ over the computed range. Under RH, the convergence $s_n(\alpha) \to 0$ for $\alpha > 1/2$ follows from the estimates of Balazard and Saias [7, Lemmes 4 et 5], but only with a bound of order $n^{-(\alpha-1/2)/3}$; the computations thus indicate a much faster decay, matching the order of the lower bound in Remark 4.2(ii). Finally, in the spirit of Weingartner's theorem, which compares the optimal coefficients with the Möbius ones in the ℓ^2-setting recalled above, we compare on the critical line the Möbius error $s_n(1/2)$ with the optimal error $d_n(1/2)^2$, the distance from E to $\operatorname{span}\{G_2, \dots, G_n\}$. While $s_n(1/2)$ keeps oscillating, $d_n(1/2)^2$ behaves like $0.046/\log n$ for $100 \le n \le 5120$, in agreement with the computations of Báez-Duarte, Balazard, Landreau and Saias [5]. Since $d_n(1/2)^2 \to 0$ under RH,

whereas $s_n(1/2) \not\to 0$ unconditionally, under RH the failure at $\alpha = 1/2$ lies in the fixed choice of coefficients $\mu(k)$ rather than in the subspaces themselves. This is in line with the work of Bettin, Conrey and Farmer [8], who showed, in the closely related setting of [5], that under RH and a hypothesis on $\sum_\rho |\zeta'(\rho)|^{-2}$ the smoothed Möbius coefficients $\mu(k)(1 - \log k/\log n)$ attain the conjectured optimal rate $(2 + \gamma - \log 4\pi)/\log n$.

The main contributions of the paper are thus: the convergence of the Möbius expansion in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$, obtained by an operator-theoretic transfer of Noor's convergence result; an exact representation of the Möbius error, free of ζ, valid for every $\alpha > 0$; and numerical evidence, up to $n = 10^6$, that for $\alpha > 1/2$ this error decays like $n^{1-2\alpha}$, much faster than the known conditional bound.

Section 2 introduces the function spaces and the results needed later. Section 3 develops the theoretical framework and proves the results described above. Section 4 presents the numerical experiments and their implications.

2 Preliminaries

We collect the function spaces and key results needed throughout the paper. The three Hilbert spaces $H^2(\mathbb{D})$, $\mathcal{H}^2$, and $H^2(\mathbb{C}_{1/2})$ are introduced here as the analytic framework within which the completeness conditions equivalent to the Riemann Hypothesis are formulated and studied.

2.1 Hardy–Hilbert spaces and Dirichlet series

Let $H^2(\mathbb{D})$ be the Hardy–Hilbert space of the unit disk, defined as

$$H^2(\mathbb{D}) = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \|f\|_{H^2}^2 = \sum_{n=0}^{\infty} |a_n|^2 < \infty \right\}.$$

A recent reformulation of the RH, introduced in [13], is stated as follows. For each $k \geq 2$, define

$$h_k(z) = \frac{1}{1-z} \log\left(\frac{1 + z + \cdots + z^{k-1}}{k} \right).$$

Theorem 2.1 (Theorem 6, [13])**.** *The Riemann Hypothesis holds if and only if the constant function* $\mathbf{1}$ *belongs to the closed linear span of* $\{h_k : k \geq 2\}$ *in* $H^2(\mathbb{D})$*.*

Let $S = M_z$ denote the shift operator on $H^2(\mathbb{D})$, corresponding to multiplication by z. An additional result from [13] plays a central role in this paper.

Lemma 2.2 (Lemma 11, [13])**.** *The series* $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I - S)h_k$ *converges to* $1 - z$ *in* $H^2(\mathbb{D})$*, where* μ *is the Möbius function.*

Let $\mathcal{H}^2$ denote the Hilbert space of Dirichlet series [12], defined as

$$\mathcal{H}^2 = \left\{ f(s) = \sum_{n=1}^{\infty} a_n n^{-s} : \|f\|_{\mathcal{H}^2}^2 = \sum_{n=1}^{\infty} |a_n|^2 < \infty \right\}.$$

Setting $H_0^2(\mathbb{D}) = H^2(\mathbb{D}) \ominus \mathbb{C}$, there exists an isometric isomorphism [12]

$$\psi : H_0^2(\mathbb{D}) \longrightarrow \mathcal{H}^2, \qquad z^k \longmapsto k^{-s}.$$

In what follows we write $H_0^2 = H_0^2(\mathbb{D})$.

2.2 The Hardy spaces $H^2(\mathbb{C}_\alpha)$

For $\alpha \in \mathbb{R}$, let $\mathbb{C}_\alpha = \{s = \sigma + it : \sigma > \alpha\}$ and let $H^2(\mathbb{C}_\alpha)$ denote the Hardy space of analytic functions F on $\mathbb{C}_\alpha$ satisfying

$$\|F\|^2_{H^2(\mathbb{C}_\alpha)} := \sup_{x>\alpha} \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(x+it)|^2 \, dt < \infty.$$

Any such F admits a non-tangential boundary value F^* almost everywhere on $\{\mathrm{Re}(s) = \alpha\}$, with

$$\|F\|^2_{H^2(\mathbb{C}_\alpha)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F^*(\alpha+it)|^2 \, dt.$$

Moreover, $H^2(\mathbb{C}_\alpha)$ is a Hilbert space with reproducing kernel

$$k_w(s) = \frac{1}{s + \overline{w} - 2\alpha}, \quad w \in \mathbb{C}_\alpha;$$

in particular, convergence in norm implies pointwise convergence in $\mathbb{C}_\alpha$. For brevity we write $\|\cdot\|_\alpha := \|\cdot\|_{H^2(\mathbb{C}_\alpha)}$.

Define the functions G_k and E in $H^2(\mathbb{C}_{1/2})$, for $k \geq 2$, as

$$G_k(s) = (k^{-s} - k^{-1})\frac{\zeta(s)}{s} \quad \text{and} \quad E(s) = \frac{1}{s}.$$

The Riemann Hypothesis reformulation for $H^2(\mathbb{C}_{1/2})$ can be stated as follows.

Theorem 2.3 (Theorem 2, [6])**.** *RH is true if and only if E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_{1/2})$.*

3 An $H^2(\mathbb{C}_\alpha)$ approximation approach

The main goal of this section is to show that E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$. For $\tau > 1/2$ and $k \geq 2$, define

$$g_{k,\tau}(z) = k^{\tau-1} \sum_{n=1}^{\infty} \frac{z^n}{n^\tau} - \sum_{n=1}^{\infty} \frac{z^{nk}}{n^\tau}.$$

It is clear that $g_{k,\tau} \in H_0^2$. Furthermore, we observe that

$$\begin{aligned}(\psi g_{k,\tau})(s) &= k^{\tau-1} \sum_{n=1}^{\infty} \frac{n^{-s}}{n^\tau} - k^{-s} \sum_{n=1}^{\infty} \frac{n^{-s}}{n^\tau} \\ &= (k^{\tau-1} - k^{-s})\zeta(s+\tau) \\ &= k^{\tau-1} f_{k,\tau}(s) \quad \text{for } \mathrm{Re}(s) > 1/2,\end{aligned}$$

where $f_{k,\tau}(s) = (1 - k^{1-s-\tau})\zeta(s+\tau)$.

3.1 A bounded operator from $\mathcal{H}^2$ to $H^2(\mathbb{C}_\alpha)$

The operator $\mathcal{M}_{1/s} : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2})$, defined by $\mathcal{M}_{1/s} : f(s) \mapsto f(s)/s$, is bounded according to [16]. For $\tau \geq 1/2$, we extend this construction by defining $\mathcal{M}^\tau : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2+\tau})$ via $\mathcal{M}^\tau : f \longmapsto f(s-\tau)/s$. To see that $\mathcal{M}^\tau$ is well defined and bounded, note that for $\sigma > \tau + 1/2$,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|f(\sigma-\tau+it)|^2}{|\sigma+it|^2}\,dt \leq \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|f(\sigma-\tau+it)|^2}{|\sigma-\tau+it|^2}\,dt \leq \|\mathcal{M}_{1/s} f\|_{1/2}^2,$$

and taking the supremum over $\sigma > \tau + 1/2$ yields

$$\|\mathcal{M}^\tau f\|_{1/2+\tau} \leq \|\mathcal{M}_{1/s}\|\,\|f\|_{\mathcal{H}^2}, \qquad \forall f \in \mathcal{H}^2.$$

The relevance of $\mathcal{M}^\tau$ to our approach lies in how it acts on the functions 1 and $f_{k,\tau}$. A direct computation gives

$$(\mathcal{M}^\tau 1)(s) = \frac{1}{s} = E(s),$$

and

$$(\mathcal{M}^\tau f_{k,\tau})(s) = \frac{f_{k,\tau}(s-\tau)}{s} = (1-k^{1-s})\frac{\zeta(s)}{s} = -kG_k(s).$$

3.2 Convergence in $H^2(\mathbb{C}_\alpha)$ for $\alpha > 1$

We now establish one of the main results of the paper. The key step is the following generalization of Lemma 2.2 to the family $\{g_{k,\tau}\}$.

Lemma 3.1. *The series $\sum_{k=2}^{\infty} \frac{\mu(k)}{k^\tau} g_{k,\tau}$ converges to $-z$ in H_0^2, for each $\tau > 1/2$.*

Proof. Since $g_{1,\tau} \equiv 0$, the $k = 1$ term may be included, giving

$$\begin{aligned}\sum_{k=2}^{n} \frac{\mu(k)}{k^\tau} g_{k,\tau}(z) &= \sum_{k=1}^{n} \frac{\mu(k)}{k^\tau}\left\{k^{\tau-1}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau}\right\} \\ &= \sum_{k=1}^{n}\frac{\mu(k)}{k}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{k=1}^{n}\frac{\mu(k)}{k^\tau}\sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau}.\end{aligned}$$

Since $\sum_{k=1}^{\infty} \mu(k)/k = 0$, a statement equivalent to the prime number theorem [1, Ch. 4], the first term vanishes in the limit:

$$\left\|\sum_{k=1}^{n}\frac{\mu(k)}{k}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau}\right\|_{H_0^2} = \left|\sum_{k=1}^{n}\frac{\mu(k)}{k}\right|\left\|\sum_{j=1}^{\infty}\frac{z^j}{j^\tau}\right\|_{H_0^2} \longrightarrow 0.$$

It therefore suffices to show that

$$\left\|-\sum_{k=1}^{n}\frac{\mu(k)}{k^\tau}\sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau} + z\right\|_{H_0^2} \longrightarrow 0. \tag{3.1}$$

A rearrangement of the double sum gives

$$\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} = \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1\le d\le n}} \mu(d) \right) + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1\le d\le n}} \mu(d) \right)$$

$$= \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left\lfloor \frac{1}{j} \right\rfloor + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j, \\ 1\le d\le n}} \mu(d) \right), \tag{3.2}$$

where the last equality uses the identity $\sum_{d|j} \mu(d) = \lfloor 1/j \rfloor$, which equals 1 for $j = 1$ and 0 for $j \ge 2$ [1, Theorem 2.1]. From (3.2),

$$\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} = z + \phi_n(z),$$

where

$$\phi_n(z) = \sum_{j=n+1}^{\infty} \left(\frac{1}{j^\tau} \sum_{\substack{d|j, \\ 1\le d\le n}} \mu(d) \right) z^j.$$

It remains to show that $\|\phi_n\|_{H_0^2} \to 0$ as $n \to \infty$. Letting $\sigma_0(j)$ denote the number of divisors of j, we estimate

$$\|\phi_n\|_{H_0^2}^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left| \sum_{\substack{d|j, \\ 1\le d\le n}} \mu(d) \right|^2 \le \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left(\sum_{d|j} 1 \right)^2 = \sum_{j=n+1}^{\infty} \frac{\sigma_0(j)^2}{j^{2\tau}}.$$

Since $\sigma_0(j) = o(j^\epsilon)$ for every $\epsilon > 0$, choosing $\epsilon > 0$ such that $\tau > 1/2 + \epsilon$ yields, for some constant $C_\epsilon > 0$,

$$\|\phi_n\|_{H_0^2}^2 \le C_\epsilon \sum_{j=n+1}^{\infty} \frac{j^{2\epsilon}}{j^{2\tau}} = C_\epsilon \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau-2\epsilon}} \longrightarrow 0,$$

since $2\tau - 2\epsilon > 1$. □

For $n \ge 2$ write $S_n := \sum_{k=2}^{n} \mu(k) G_k$ for the n-th Möbius partial sum. With this notation, Lemma 3.1 transfers to the half-plane setting as follows.

Theorem 3.2. *The sequence (S_n) converges to E in $H^2(\mathbb{C}_{1/2+\tau})$, for every $\tau > 1/2$.*

Proof. By Lemma 3.1 and the isomorphism ψ,

$$\sum_{k=2}^{n} \frac{\mu(k)}{k} f_{k,\tau} \longrightarrow -1 \quad \text{in } \mathcal{H}^2.$$

Applying $\mathcal{M}^\tau$ and using $(\mathcal{M}^\tau f_{k,\tau})(s) = -kG_k(s)$, we conclude

$$S_n = \sum_{k=2}^{n} \mu(k) G_k \longrightarrow E \quad \text{in } H^2(\mathbb{C}_{1/2+\tau}). \tag{3.3}$$

□

Corollary 3.3. *E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$.*

The membership of E in the closed linear span of $\{G_k : k \geq 2\}$ also has direct consequences for the location of the zeros of ζ.

Proposition 3.4. *For any $\alpha \geq 1/2$,*

$$E \in \overline{\text{span}}\{G_k : k \geq 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \neq 0, \quad \forall s \in \mathbb{C}_\alpha.$$

Proof. Let ρ be a zero of ζ in $\mathbb{C}_\alpha$. Then $G_k(\rho) = 0$ for every $k \geq 2$. Since $H^2(\mathbb{C}_\alpha)$ is a reproducing kernel Hilbert space, convergence in norm implies pointwise convergence, so $E(\rho) = 1/\rho = 0$, which is impossible. □

3.3 A lower bound from the zeros of ζ

Proposition 3.4 is qualitative: it turns convergence into a zero-free half-plane. The reproducing kernel of $H^2(\mathbb{C}_\alpha)$ yields a quantitative companion in the opposite direction, in which a single zero of ζ inside $\mathbb{C}_\alpha$ produces an explicit, n-independent floor for the approximation error. For $n \geq 2$ and $\alpha > 0$, define the *optimal approximation error*

$$d_n(\alpha)^2 := \inf_{c_2,\dots,c_n \in \mathbb{C}} \Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2,$$

the best approximation of E by the first $n-1$ generators. The restriction $\alpha > 0$ is precisely what is needed for E and the G_k to belong to $H^2(\mathbb{C}_\alpha)$: they are Mellin transforms of bounded functions supported in $(0, 1]$, and therefore belong to $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 0$ (see the proof of Proposition 4.1), whereas for $\alpha \leq 0$ they have a pole at $s = 0$. Taking $c_k = \mu(k)$ shows $d_n(\alpha)^2 \leq \|S_n - E\|_\alpha^2$, so any lower bound for $d_n(\alpha)^2$ is a fortiori one for the Möbius error.

Theorem 3.5. *Let $\alpha > 0$ and suppose ζ has a zero ρ with $\operatorname{Re}(\rho) > \alpha$. Then for every $n \geq 2$ and all coefficients $c_2, \dots, c_n \in \mathbb{C}$,*

$$\Big\| E - \sum_{k=2}^{n} c_k G_k \Big\|_\alpha^2 \;\geq\; \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2} \;=\; \frac{2\,(\operatorname{Re}\rho - \alpha)}{|\rho|^2} \;>\; 0.$$

More generally, if $\rho_1, \dots, \rho_m$ are distinct zeros of ζ in $\mathbb{C}_\alpha$, then for every $n \geq 2$,

$$d_n(\alpha)^2 \;\geq\; v^* K^{-1} v, \qquad K = \Big(\tfrac{1}{\rho_i + \overline{\rho_j} - 2\alpha} \Big)_{i,j=1}^{m}, \qquad v = \big(1/\rho_i\big)_{i=1}^{m}.$$

In particular $d_n(\alpha)^2$ is bounded below by a positive constant independent of n, so $d_n(\alpha)^2 \not\to 0$.

Proof. Since $\mathrm{Re}(\rho) > \alpha$ we have $\rho \in \mathbb{C}_\alpha$, and $\zeta(\rho) = 0$ gives $G_k(\rho) = (k^{-\rho} - k^{-1})\zeta(\rho)/\rho = 0$ for every k. Hence $F := E - \sum_{k=2}^{n} c_k G_k$ satisfies $F(\rho) = E(\rho) = 1/\rho$. By the reproducing property $F(\rho) = \langle F, k_\rho \rangle_\alpha$, so the Cauchy–Schwarz inequality gives

$$\|F\|_\alpha^2 \;\geq\; \frac{|\langle F, k_\rho\rangle_\alpha|^2}{\|k_\rho\|_\alpha^2} = \frac{|E(\rho)|^2}{\|k_\rho\|_\alpha^2}.$$

With $\|k_\rho\|_\alpha^2 = k_\rho(\rho) = (2\,\mathrm{Re}\,\rho - 2\alpha)^{-1}$ and $|E(\rho)|^2 = |\rho|^{-2}$, this is the first inequality.

For the general case, let $N = \{g \in H^2(\mathbb{C}_\alpha) : g(\rho_i) = 0,\ 1 \leq i \leq m\}$, whose orthogonal complement is $V = \mathrm{span}\{k_{\rho_1}, \ldots, k_{\rho_m}\}$. Every combination $\sum_k c_k G_k$ lies in N, since $G_k(\rho_i) = 0$; thus, with P the orthogonal projection onto V,

$$\Big\|E - \sum_k c_k G_k\Big\|_\alpha \;\geq\; \Big\|P\big(E - \textstyle\sum_k c_k G_k\big)\Big\|_\alpha = \|PE\|_\alpha.$$

A standard computation with the kernels gives $\|PE\|_\alpha^2 = v^* K^{-1} v$, where $K_{ij} = \langle k_{\rho_j}, k_{\rho_i}\rangle_\alpha = k_{\rho_j}(\rho_i) = (\rho_i + \overline{\rho_j} - 2\alpha)^{-1}$ and $v_i = \langle E, k_{\rho_i}\rangle_\alpha = E(\rho_i) = 1/\rho_i$. Taking the infimum over c yields the bound on $d_n(\alpha)^2$; the right-hand side is a fixed positive constant, so $d_n(\alpha)^2 \not\to 0$. □

Corollary 3.6. *Let* $\Theta = \sup\{\mathrm{Re}\,\rho : \rho \text{ a non-trivial zero of } \zeta\}$ *and* $\alpha_c = \inf\big\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\big\}$. *Then* $\tfrac{1}{2} \leq \Theta \leq \alpha_c \leq 1$, *and* $\alpha_c = \tfrac{1}{2}$ *if and only if RH holds.*

Proof. By Theorem 3.2 the set $\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\}$ is nonempty and bounded below by $\tfrac{1}{2}$, so α_c is well-defined and $\alpha_c \leq 1$. For $\alpha < \Theta$ pick a zero ρ with $\mathrm{Re}\,\rho > \alpha$; Theorem 3.5 gives $d_n(\alpha)^2 \not\to 0$, whence $\Theta \leq \alpha_c$. As $\Theta \geq \tfrac{1}{2}$ is classical, $\tfrac{1}{2} \leq \Theta \leq \alpha_c \leq 1$. If $\alpha_c = \tfrac{1}{2}$ then $\Theta = \tfrac{1}{2}$, which is RH. Conversely, under RH, Theorem 2.3 places E in the closed linear span of $\{G_k\}_{k\geq 2}$ in $H^2(\mathbb{C}_{1/2})$. Since $d_n(1/2)$ is non-increasing in n, this forces $d_n(1/2)^2 \to 0$; hence $\alpha = \tfrac{1}{2}$ belongs to the set $\{\alpha \geq \tfrac{1}{2} : d_n(\alpha)^2 \to 0\}$, so that $\alpha_c \leq \tfrac{1}{2}$. Combined with $\Theta \geq \tfrac{1}{2}$, this gives $\alpha_c = \tfrac{1}{2}$. □

4 Numerical experiments

The range $1/2 \leq \alpha \leq 1$ is not covered by Theorem 3.2; under RH, convergence there follows from the estimates of Balazard and Saias [7] (see the end of Subsection 4.2), but only with a weak bound. This section probes this range numerically. Two questions guide the experiments: how the Möbius error $s_n(\alpha) = \|S_n - E\|_\alpha^2$ behaves as $n \to \infty$ across the strip, and whether the obstruction on the critical line traced by Corollary 3.6 is intrinsic to the subspaces $\mathrm{span}\{G_2, \ldots, G_n\}$ or only to the fixed choice of coefficients $\mu(k)$. The first question is addressed in Subsection 4.2 and the second in Subsection 4.3. Both rest on the computation of norms in $H^2(\mathbb{C}_\alpha)$, which we describe first.

4.1 Computation of the norms

A direct evaluation of the boundary integral

$$s_n(\alpha) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \big|S_n(\alpha + it) - E(\alpha + it)\big|^2\, dt$$

by numerical quadrature is costly and unreliable for large n: every evaluation of the integrand involves $\zeta(\alpha+it)$ and the $n-1$ oscillating factors $k^{-\alpha-it}$, and the integrand oscillates on scales that shrink as n and $|t|$ grow. We avoid quadrature altogether by means of the following exact representation of $s_n(\alpha)$ as a series with elementary terms.

Proposition 4.1. *Let $\alpha > 0$ and $n \geq 2$. For $m \geq 1$ put*

$$R_n(m) = 1 + \sum_{k=2}^{n} \mu(k)\, \frac{m \bmod k}{k} \qquad \text{and} \qquad w_m(\alpha) = \int_m^{m+1} \frac{du}{u^{2\alpha+1}} = \frac{m^{-2\alpha} - (m+1)^{-2\alpha}}{2\alpha}.$$

Then

$$s_n(\alpha) = \sum_{m=1}^{\infty} w_m(\alpha)\, R_n(m)^2. \tag{4.1}$$

Proof. Write $\{y\}$ for the fractional part of y and, for $k \geq 2$, let $\rho_k(x) = \{1/(kx)\} - \{1/x\}/k$ for $x > 0$. If $x > 1$, both $1/(kx)$ and $1/x$ lie in $(0,1)$, so $\rho_k(x) = 0$; moreover $|\rho_k| \leq 2$. For $0 < \operatorname{Re}(s) < 1$ we have the classical formula [18, §2.1], written here after the change of variables $x \mapsto 1/x$,

$$\int_0^{\infty} \{1/x\}\, x^{s-1}\, dx = -\frac{\zeta(s)}{s},$$

and the substitution $y = kx$ gives $\int_0^{\infty} \{1/(kx)\}\, x^{s-1}\, dx = -k^{-s}\zeta(s)/s$. Subtracting the two identities,

$$\int_0^1 \rho_k(x)\, x^{s-1}\, dx = -\bigl(k^{-s} - k^{-1}\bigr)\frac{\zeta(s)}{s} = -G_k(s), \qquad 0 < \operatorname{Re}(s) < 1.$$

Since ρ_k is bounded on $(0,1)$, the left-hand side converges absolutely and is analytic on $\operatorname{Re}(s) > 0$, so by analytic continuation the identity holds on the whole half-plane $\mathbb{C}_0$. Together with $E(s) = \int_0^1 x^{s-1}\, dx$ this gives

$$E(s) - S_n(s) = \int_0^1 f_n(x)\, x^{s-1}\, dx, \qquad f_n = 1 + \sum_{k=2}^{n} \mu(k)\, \rho_k, \qquad \operatorname{Re}(s) > 0.$$

Let $\sigma \geq \alpha$. On the line $\operatorname{Re}(s) = \sigma$ the substitution $x = e^{-y}$ turns this integral into the Fourier transform of $g_\sigma(y) = f_n(e^{-y})\, e^{-\sigma y}$, $y > 0$, which lies in $L^1(0,\infty) \cap L^2(0,\infty)$ because f_n is bounded and $\sigma > 0$. By Plancherel's theorem [17, Theorem 9.13],

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \bigl|(E - S_n)(\sigma + it)\bigr|^2\, dt = \int_0^{\infty} |g_\sigma(y)|^2\, dy = \int_0^1 |f_n(x)|^2\, x^{2\sigma-1}\, dx.$$

Since $0 < x < 1$, the right-hand side is a non-increasing function of σ, and it tends to its value at $\sigma = \alpha$ as $\sigma \to \alpha^+$ by monotone convergence. Hence $E - S_n \in H^2(\mathbb{C}_\alpha)$ and, by the definition of the norm,

$$s_n(\alpha) = \sup_{\sigma > \alpha} \int_0^1 |f_n(x)|^2\, x^{2\sigma-1}\, dx = \int_0^1 |f_n(x)|^2\, x^{2\alpha-1}\, dx.$$

Finally put $x = 1/u$, so that the last integral equals $\int_1^\infty |f_n(1/u)|^2\, u^{-2\alpha-1}\, du$. For $u \in [m, m+1)$ the interval $(m, u]$ contains no integer, hence no multiple of k; thus $\{u/k\} = u/k - \lfloor m/k \rfloor$ and $\{u\} = u - m$, which give $\rho_k(1/u) = m/k - \lfloor m/k \rfloor = (m \bmod k)/k$ and $f_n(1/u) = R_n(m)$. Splitting the integral over the intervals $[m, m+1)$ yields (4.1). □

Remark 4.2. (i) For $\alpha = 1/2$ the weights are $w_m(1/2) = 1/(m(m+1))$, and Proposition 4.1 turns the approximation of E in $H^2(\mathbb{C}_{1/2})$ into an approximation problem for the constant sequence $\mathbf{1}$ by the sequences r_k/k, $r_k(m) = m \bmod k$, in a weighted ℓ^2-space; compare with the setting of Balazard and Saias [7] recalled in the introduction. The proof also shows that $s_n(1/2) = \|f_n\|^2_{L^2(0,1)}$, which makes explicit the link with the setting of Báez-Duarte. Indeed, put $\chi = \chi_{(0,1]}$, $M_1(n) = \sum_{k \le n} \mu(k)/k$, and extend f_n by 0 to $(1, \infty)$. Since $\sum_{k=2}^n \mu(k)/k = M_1(n) - 1$, the definition of the ρ_k gives

$$f_n(x) = \chi(x) + \sum_{j=1}^{n} \mu(j) \Big\{ \frac{1}{jx} \Big\} - M_1(n) \Big\{ \frac{1}{x} \Big\}, \qquad x > 0,$$

both sides vanishing for $x > 1$, where the right-hand side equals $M_1(n)/x - M_1(n)/x$. As $\{1/x\} \in L^2(0, \infty)$ and $M_1(n) \to 0$, the convergence $s_n(1/2) \to 0$ would force the partial sums of (1.1) to converge to χ in $L^2(0, \infty)$, contradicting [3, Theorem 2.2]. Hence $s_n(1/2) \not\to 0$; and since, by Theorem 3.2, any limit of (S_n) in $H^2(\mathbb{C}_{1/2})$ would coincide with E on $\mathbb{C}_1$, hence on $\mathbb{C}_{1/2}$, the sequence (S_n) does not converge in $H^2(\mathbb{C}_{1/2})$.

(ii) Since $\sum_{k \le m} \mu(k) \lfloor m/k \rfloor = 1$ for every $m \ge 1$ [1, Theorem 3.12], and $\lfloor m/k \rfloor = 0$ for $k > m$, for $m \le n$ we get

$$R_n(m) = 1 + \sum_{k=1}^{n} \mu(k) \Big(\frac{m}{k} - \Big\lfloor \frac{m}{k} \Big\rfloor \Big) = m\, M_1(n), \qquad M_1(n) = \sum_{k \le n} \frac{\mu(k)}{k},$$

and therefore

$$s_n(\alpha) \ge M_1(n)^2 \sum_{m=1}^{n} m^2\, w_m(\alpha), \qquad \sum_{m=1}^{n} m^2\, w_m(\alpha) \sim \frac{n^{2-2\alpha}}{2-2\alpha} \quad \Big(\tfrac{1}{2} \le \alpha < 1 \Big). \tag{4.2}$$

Thus the decay of $s_n(\alpha)$ is limited from below by the size of the partial sums $M_1(n)$ of $\sum_k \mu(k)/k$. Under RH one has $M_0(x) = \sum_{k \le x} \mu(k) = O(x^{1/2+\varepsilon})$ [18, Theorem 14.25(C)]; since $\sum_k \mu(k)/k = 0$, partial summation gives $M_1(x) = M_0(x)/x - \int_x^\infty M_0(t)\, t^{-2}\, dt$, whence $M_1(n) = O(n^{-1/2+\varepsilon})$; if $M_1(n)$ is of exact order $n^{-1/2}$ along a sequence of values of n, the right-hand side of (4.2) is of order $n^{1-2\alpha}$ along that sequence.

Implementation. We evaluate the series (4.1) in three parts. The terms with $m \le m_0$ are summed directly. On $m_0 < m \le M$ the terms are grouped into blocks $[a, b)$ of relative length $(b-a)/a \le h$, and the weight is expanded to second order about the centre c of each block,

$$\sum_{a \le m < b} w_m(\alpha)\, R_n(m)^2 \approx \sum_{j=0}^{2} \frac{w^{(j)}(c)}{j!} \sum_{a \le m < b} (m-c)^j R_n(m)^2, \qquad w(x) = \frac{x^{-2\alpha} - (x+1)^{-2\alpha}}{2\alpha},$$

which has relative error $O(h^3)$ and reduces the dependence on α to three moments of R_n^2 per block, shared by all values of α. For $m > M$ we replace $R_n(m)^2$ by its mean value $\overline{R_n^2}$ over a period of $m \mapsto R_n(m)$, so that the tail contributes $\overline{R_n^2}\,(M+1)^{-2\alpha}/(2\alpha)$. This replacement is not exact, since the period $\operatorname{lcm}(2,\dots,n)$ of $m \mapsto R_n(m)$ is far larger than M; its effect is part of the truncation error estimated below. The mean itself is known exactly. Let $g = \gcd(j,k)$. For m uniformly distributed modulo $\operatorname{lcm}(j,k)$, the Chinese remainder theorem gives $m \bmod j = ga + r$ and $m \bmod k = gb + r$, where $r = m \bmod g$, $a \in \{0,\dots,j/g-1\}$ and $b \in \{0,\dots,k/g-1\}$ are independent and uniformly distributed; hence the covariance of $m \bmod j$ and $m \bmod k$ equals the variance of r, namely $(g^2-1)/12$. Writing $\gcd(j,k)^2 = \sum_{d\mid\gcd(j,k)} J_2(d)$, where $J_2(d) = d^2\prod_{p\mid d}(1-p^{-2})$ is Jordan's totient, characterised by $\sum_{d\mid q} J_2(d) = q^2$ for $q \ge 1$ [1, Ch. 2, Exercise 17] (the term $d=1$ cancels the -1), one finds

$$\overline{R_n^2} = \Big(1 + \sum_{k=2}^{n} \mu(k)\,\frac{k-1}{2k}\Big)^2 + \frac{1}{12}\sum_{d\ge 2} J_2(d)\Big(\sum_{\substack{2\le k\le n\\ d\mid k}} \frac{\mu(k)}{k}\Big)^2.$$

Since $R_n(m) = R_{n-1}(m) + \mu(n)\,(m \bmod n)/n$, passing from $n-1$ to n costs a single pass over $m \le M$, and only the n with $\mu(n) \ne 0$ contribute. All the values $s_n(\alpha)$ with $n \le N$ are therefore obtained with $O(NM)$ elementary operations, without evaluating ζ and without numerical quadrature. Since all terms of (4.1) are non-negative, no cancellation occurs when the series is summed. All computations are carried out in IEEE double precision, without compensated summation; two implementations of the recursion that accumulate $R_n(m)$ in different orders agree to within $5\cdot 10^{-9}$ (relative), far below the truncation error estimated below. The computation was organised so that each block of values of m stays in cache while it runs through all n, the blocks being processed in parallel. A Fortran 2008 program with OpenMP computed $s_n(\alpha)$ for the 51 values $\alpha = 0.50, 0.51, \dots, 1.00$ and the 3001 logarithmically spaced integers n_j, $0 \le j \le 3000$, where n_j is the integer nearest to $10^{3+j/1000}$ (these integers are distinct, and $n_{3000} = 10^6$), in 40 minutes on the machine described at the end of this subsection, with $m_0 = 10^4$, $h = 10^{-3}$ and $M = 3\cdot 10^8$. The block expansion changes the results by less than 10^{-9} (relative) with respect to direct summation. To control the truncation error we repeated the whole computation with $M = 10^8$ and with $M = 10^9$. With respect to the run with $M = 10^9$, the values obtained with $M = 3\cdot 10^8$ differ by at most 10^{-8} for $n < 10^4$, $7.4\cdot 10^{-7}$ for $10^4 \le n < 10^5$ and $4.8\cdot 10^{-5}$ for $10^5 \le n \le 10^6$, while those obtained with $M = 10^8$ differ from the run with $M = 3\cdot 10^8$ by up to $3.5\cdot 10^{-4}$. These differences are consistent with a truncation error decreasing like $M^{-1.8}$, so that the relative error of the values reported below, computed with $M = 3\cdot 10^8$, is about $5\cdot 10^{-5}$ at most; the entries of Table 1 change by at most $1.1\cdot 10^{-5}$ when $M = 10^9$ is used instead. As an independent check, the values of $s_n(1/2)$ given by (4.1) agree with those obtained from the Gram data described below, which are computed from the boundary integral and the zeta function, to within $3.4\cdot 10^{-4}$ (relative) for $2 \le n \le 5120$, the median difference being $6\cdot 10^{-5}$. The same check on the line $\operatorname{Re}(s) = 3/4$, with the Gram data computed there in the same way, gives agreement with (4.1) to within 10^{-4} for $2 \le n \le 5120$, the median difference being 10^{-5}.

Optimal error. The optimal error of Subsection 3.3 is computed on the critical line as $d_n(1/2)^2 = \|E\|_{1/2}^2 - b^{\top}A^{-1}b$, with Gram data $A = (\langle G_j, G_k\rangle_{1/2})_{j,k=2}^{n}$ and $b = (\langle E, G_k\rangle_{1/2})_{k=2}^{n}$.

Because E and the G_k are real on the real axis, they satisfy the reflection symmetry $G_k(\bar{s}) = \overline{G_k(s)}$ (and likewise for E), so each such inner product reduces to a real half-line integral,

$$\langle G_j, G_k\rangle_{1/2} = \frac{1}{\pi}\int_0^\infty \operatorname{Re}\left[G_j(\tfrac12+it)\,\overline{G_k(\tfrac12+it)}\right]dt,$$

and likewise for $\langle E, G_k\rangle_{1/2}$; hence A and b are real and the minimising coefficients may be taken real. These boundary integrals are evaluated with the trapezoidal rule on a fixed grid of 33 600 nodes, 8000 of them equally spaced on $[0, 80)$ and 25 600 geometrically spaced on $[80, 3\cdot 10^6]$, with $\zeta(\frac12+it)$ computed by the Euler–Maclaurin formula with $N \ge 2|s|/\pi$ terms and twelve Bernoulli corrections, whose remainder is negligible in double precision. On this grid $\|E\|_{1/2}^2 = 1$ is reproduced to $1.1\cdot 10^{-7}$, the values of $\langle E, G_k\rangle_{1/2}$ agree to within 10^{-4} (relative) with the exact value $G_k(1) = -(\log k)/k$, which holds because E is the reproducing kernel of $H^2(\mathbb{C}_{1/2})$ at $s = 1$, and the values of $s_n(1/2)$ computed from the Gram data agree with (4.1) as stated above; these checks control the accuracy of the inner products. The Gram matrix for $n = 5120$ is factorised as $A = U^\top U$ (Cholesky), and with $y = U^{-\top}b$ the leading blocks of the factorisation give

$$d_n(1/2)^2 = \|E\|_{1/2}^2 - \sum_{i=1}^{n-1} y_i^2, \qquad 2 \le n \le 5120,$$

for all n at once. Since A is ill-conditioned, the accuracy of the inner products does not by itself bound the error of $d_n(1/2)^2$. We therefore also computed $d_n(1/2)^2$ for $n \le 1280$ from the exact Gram data that follow from the proof of Proposition 4.1,

$$\langle G_j, G_k\rangle_{1/2} = \frac{1}{jk}\sum_{m=1}^{\infty}\frac{(m \bmod j)(m \bmod k)}{m(m+1)}, \qquad \langle E, G_k\rangle_{1/2} = -\frac{\log k}{k},$$

where the series is summed exactly over one period $L = \operatorname{lcm}(j,k)$ by means of $\sum_{q\ge0}\big((qL+r)(qL+r+1)\big)^{-1} = \big(\psi((r+1)/L) - \psi(r/L)\big)/L$, ψ being the digamma function; closed formulas for these inner products were obtained by Vasyunin [19]. The two computations of $d_n(1/2)^2$ agree to within $2\cdot 10^{-4}$ (relative) for $n < 640$ and to within $5\cdot 10^{-4}$ for $640 \le n \le 1280$, the difference growing slowly with n as A becomes more ill-conditioned; for larger n the comparison with [5] in Subsection 4.3 provides a further check. The exact Gram data also give $s_n(1/2)$, which agrees with the implementation of (4.1) described above to within $4\cdot 10^{-8}$.

Computing environment. All the computations of this section were carried out with programs written in Fortran 2008 and parallelised with OpenMP, on a desktop computer with an Intel Core i7-13700KF processor (16 cores, 8 performance and 8 efficiency cores, 24 hardware threads, maximum clock frequency 5.4 GHz, 30 MB of L3 cache) and 32 GB of RAM, running Ubuntu 26.04 LTS (Linux kernel 7.0). The programs were compiled with GNU Fortran 15.2 with the options `-O3 -march=native -fopenmp` and run with 20 OpenMP threads; the matrix products and the Cholesky factorisation of the optimal-error computation use the BLAS and LAPACK libraries. The computation of $d_n(1/2)^2$ for $2 \le n \le 5120$ took less than two minutes. The figures were produced with Python (Matplotlib) from the output of the Fortran programs. The code generating all figures and tables of this section is available from the authors upon request.

4.2 The Möbius error across the strip

We computed $s_n(\alpha)$ for $\alpha = 0.50, 0.51, \ldots, 1.00$ and $10^3 \le n \le 10^6$, at 1000 logarithmically spaced values of n per decade. Figure 1 shows the sequences for $\alpha = 0.50, 0.60, \ldots, 1.00$. They oscillate strongly with n: within each decade the largest value exceeds the smallest by a factor of about 8 on the critical line, and by a factor of several hundred for $\alpha = 1$. The oscillations occur at the same values of n for all α, and the value of $s_n(\alpha)$ at an isolated n is therefore not representative of its behaviour. We summarise each sequence instead by its largest and smallest values over each decade of n (Table 1), and measure its decay by the change of $\log_{10} \max s_n$ and $\log_{10} \min s_n$ per decade between the first and the last decade (Figure 2). Since it is based on extreme values, this two-point estimate is only indicative.

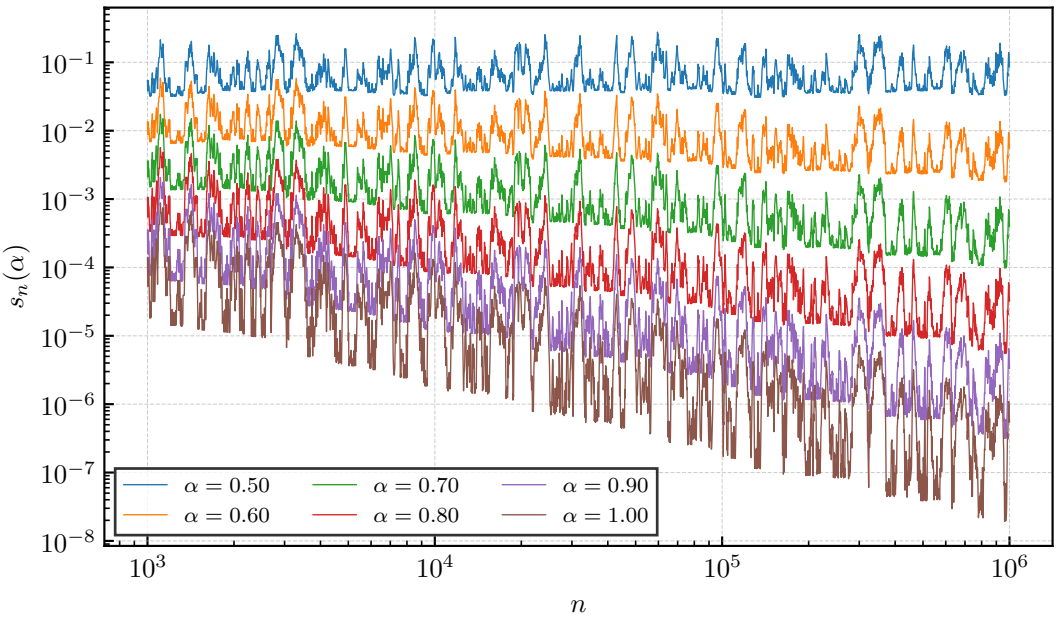


Figure 1: The Möbius error $s_n(\alpha)$ for $\alpha = 0.50, 0.60, \ldots, 1.00$ and $10^3 \le n \le 10^6$, in log–log scale.

On the critical line the Möbius error does not decay: over the three decades the maxima of $s_n(1/2)$ are 0.262, 0.277 and 0.253, and the minima $3.18 \cdot 10^{-2}$, $3.41 \cdot 10^{-2}$ and $3.08 \cdot 10^{-2}$. This is consistent with the result of Báez-Duarte [3], by which $s_n(1/2) \not\to 0$ (Remark 4.2). For every $\alpha > 1/2$, on the other hand, both the maxima and the minima decrease geometrically from one decade to the next, and Figure 2 shows that the rate is very close to $10^{1-2\alpha}$ per decade: over the whole range the observed changes of $\log_{10}$ per decade differ from $1 - 2\alpha$ by less than 0.08. The data are thus consistent with

$$s_n(\alpha) \asymp n^{1-2\alpha}, \qquad \tfrac{1}{2} \le \alpha < 1,$$

for the envelope of the oscillations (for $\alpha = 1$ see below), and in particular with $s_n(\alpha) \to 0$ for every $\alpha > 1/2$. The convergence is very slow near the critical line: for $\alpha = 0.55$ the error

α	$10^3 \le n < 10^4$		$10^4 \le n < 10^5$		$10^5 \le n \le 10^6$	
	$\max s_n$	$\min s_n$	$\max s_n$	$\min s_n$	$\max s_n$	$\min s_n$
0.50	2.617e−1	3.181e−2	2.772e−1	3.408e−2	2.530e−1	3.079e−2
0.55	1.219e−1	1.189e−2	9.722e−2	1.086e−2	7.554e−2	7.605e−3
0.60	5.866e−2	4.390e−3	3.954e−2	3.213e−3	2.298e−2	1.776e−3
0.65	3.162e−2	1.632e−3	1.683e−2	9.559e−4	7.140e−3	4.171e−4
0.70	1.740e−2	6.106e−4	7.338e−3	2.857e−4	2.393e−3	9.806e−5
0.75	9.798e−3	2.296e−4	3.291e−3	8.578e−5	8.578e−4	2.316e−5
0.80	5.662e−3	8.677e−5	1.528e−3	2.584e−5	3.197e−4	5.499e−6
0.85	3.370e−3	3.293e−5	7.396e−4	7.811e−6	1.254e−4	1.313e−6
0.90	2.073e−3	1.255e−5	3.768e−4	2.368e−6	5.263e−5	3.157e−7
0.95	1.323e−3	4.793e−6	2.039e−4	7.196e−7	2.405e−5	7.693e−8
1.00	8.784e−4	1.838e−6	1.181e−4	2.192e−7	1.217e−5	1.926e−8

Table 1: Largest and smallest values of the Möbius error $s_n(\alpha)$ over each decade of n, for $\alpha = 0.50, 0.55, \ldots, 1.00$, computed at 3001 logarithmically spaced values of $n \in [10^3, 10^6]$. The maxima and minima are taken over these sampled values of n.

decreases only by a factor $10^{0.1} \approx 1.26$ per decade, so that for moderate n the behaviour for α slightly above $1/2$ is hard to tell apart from that on the critical line. The exponent $1 - 2\alpha$ is precisely the one given by the lower bound (4.2) when $M_1(n)$ is of order $n^{-1/2}$, and the largest peaks of $s_n(\alpha)$ in Figure 1 occur where $nM_1(n)^2$ is large. Near $\alpha = 1$ the maxima decay more slowly than $n^{1-2\alpha}$ (Figure 2). This is consistent with (4.2): for $\alpha = 1$ the sum $\sum_{m \le n} m^2 w_m(1)$ grows like $\log n$, so that along a sequence with $M_1(n)$ of exact order $n^{-1/2}$ the lower bound is of order $n^{-1} \log n$, whose slope per decade over the computed range is about -0.91, close to the observed -0.93.

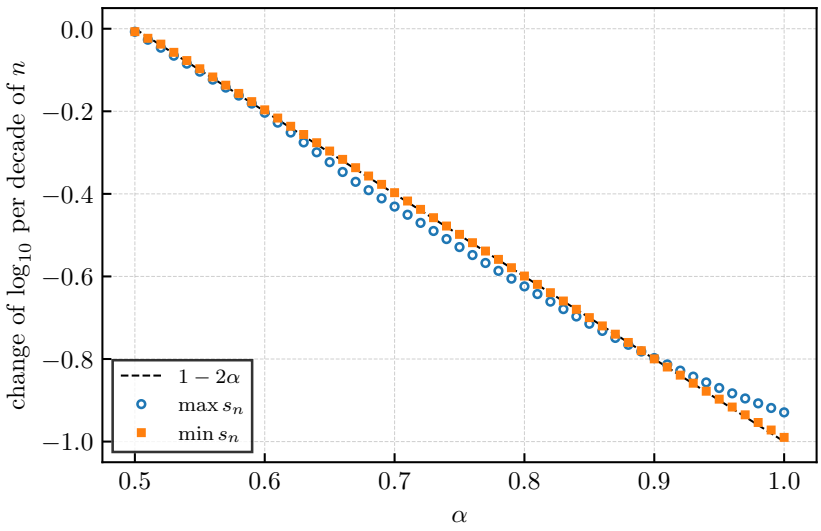


Figure 2: Change of $\log_{10} \max s_n(\alpha)$ and of $\log_{10} \min s_n(\alpha)$ per decade of n, from $10^3 \le n < 10^4$ to $10^5 \le n \le 10^6$, for $\alpha = 0.50, 0.51, \ldots, 1.00$, compared with $1 - 2\alpha$ (dashed).

These observations must be read with care. By Proposition 3.4, a proof that $s_n(\alpha) \to 0$ for some $\alpha < 1$ would give a zero-free half-plane $\mathbb{C}_\alpha$, far beyond the known zero-free regions of ζ. The behaviour observed is the one expected under RH. Indeed, by Remark 4.2, $f_n = 1 + S_n^* - M_1(n)\{1/x\}$ on $(0,1]$ with $S_n^*(x) = \sum_{k\le n} \mu(k)\{1/(kx)\}$, and if RH holds the estimates of Balazard and Saias [7, Lemmes 4 et 5] give $\int_0^1 |1 + S_n^*(x)|^2 x^{2\alpha-1}\,dx \ll_\alpha n^{-(\alpha-1/2)/3}$ for $1/2 < \alpha \le 1$; together with $M_1(n) = O(n^{-1/2+\varepsilon})$, this yields $s_n(\alpha) = O(n^{-(\alpha-1/2)/3})$, and in particular $s_n(\alpha) \to 0$, for every $\alpha > 1/2$. The decay like $n^{1-2\alpha}$ observed above is much faster, and matches the order of the lower bound (4.2). Nevertheless, no finite computation can rule out the alternative: by Theorem 3.5, a zero ρ with $\operatorname{Re}\rho > \alpha$ only forces $s_n(\alpha) \ge 2(\operatorname{Re}\rho - \alpha)/|\rho|^2$, and since every zero with $|\operatorname{Im}\rho| \le 3 \cdot 10^{12}$ lies on the critical line [15], this floor would be smaller than 10^{-24}.

4.3 The optimal error on the critical line

The critical line is the one place in the range studied where S_n demonstrably fails: by Báez-Duarte [3] the Möbius coefficients do not sum to E at $\alpha = 1/2$ (Remark 4.2), and the undamped oscillations of Figure 1 are the price of this fixed choice. Since it is exactly at $\alpha = 1/2$ that Theorem 2.3 formulates the RH, the natural question is what this failure rules out. The answer is: nothing about the subspaces $\operatorname{span}\{G_2, \dots, G_n\}$ themselves, since $\mu(k)$ is only one admissible choice of coefficients, and it is the subspaces, not the choice, that Theorem 3.5 ties to the zeros of ζ. This is what motivates the present experiment: we replace $s_n(1/2)$ by the optimal error $d_n(1/2)^2$ of Subsection 3.3, the true distance from E to $\operatorname{span}\{G_2, \dots, G_n\}$ on the critical line, which removes the coefficients from the picture by optimising over all of them. It is computed by the Gram matrix method of Subsection 4.1.

Figure 3 contrasts the two sequences, and Table 2 records their values at selected n. While the Möbius error $s_n(1/2)$ keeps oscillating between about $3 \cdot 10^{-2}$ and $3 \cdot 10^{-1}$, the optimal error $d_n(1/2)^2$, which is non-increasing in n by definition, decreases slowly: for $100 \le n \le 5120$, $d_n(1/2)^2 \log n$ stays between 0.044 and 0.048. Under RH, $d_n(1/2)^2 \to 0$ by Theorem 2.3, whereas $s_n(1/2) \not\to 0$ unconditionally (Remark 4.2); thus, under RH, the failure of S_n on the critical line is due to the fixed choice $c_k = \mu(k)$ and not to the subspaces $\operatorname{span}\{G_2, \dots, G_n\}$. Our computations illustrate this contrast; they cannot, of course, decide whether $d_n(1/2)^2 \to 0$ (see the end of Subsection 4.2).

The distance $d_n(1/2)$ has been studied before in the Nyman–Beurling setting. By the proof of Proposition 4.1, $d_n(1/2)$ is the distance in $L^2(0,\infty)$ from $\chi = \chi_{(0,1]}$ to the functions $\sum_{k=1}^n c_k\{1/(kx)\}$ with $\sum_k c_k/k = 0$. Dropping this constraint gives the distance studied by Báez-Duarte, Balazard, Landreau and Saias [5], which is in turn bounded below by their $D(1/n)$. Hence, by [5, Théorème] as sharpened by Burnol [10, Theorem 1.3],

$$\liminf_{n\to\infty} d_n(1/2)^2 \log n \;\ge\; 2 + \gamma - \log 4\pi = 0.0461\ldots$$

unconditionally: if RH fails the bound is trivial, since then $d_n(1/2)$ does not tend to 0, while under RH Burnol's bound $\sum_\rho m_\rho^2/|\rho|^2$, where m_ρ is the multiplicity of the zero ρ, is at least $\sum_\rho m_\rho/|\rho|^2 = 2 + \gamma - \log 4\pi$. Being a bound on the $\liminf$, it is compatible with the values of $d_n(1/2)^2 \log n$ slightly below $2 + \gamma - \log 4\pi$ found above for finite n. In [5, §8] the unconstrained

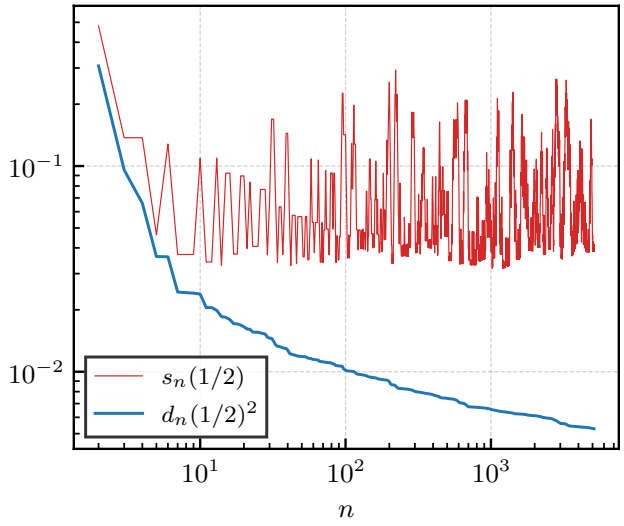


Figure 3: The optimal error $d_n(1/2)^2$ (blue) and the Möbius error $s_n(1/2)$ (red) on the critical line, $2 \le n \le 5120$, in log–log scale.

n	$d_n(1/2)^2$	$s_n(1/2)$
40	1.274e−2	1.442e−1
160	9.357e−3	6.847e−2
640	7.023e−3	3.659e−2
1280	6.366e−3	3.474e−2
2560	5.932e−3	4.314e−2
5120	5.283e−3	3.851e−2

Table 2: The same two errors at selected n; for $n \le 1280$ the values of $d_n(1/2)^2$ are those of the exact Gram data.

distance was computed for $n \le 10^4$ from Vasyunin's closed formulas for the inner products [19], which led them to conjecture [5, Conjecture 2] that $d_n^2 \sim (2 + \gamma - \log 4\pi)/\log n$. Our values, obtained by a different method (the exact Gram data of Subsection 4.1 for $n \le 1280$ and quadrature on the critical line beyond), are consistent with these results, although the unconstrained distance of [5] does not exceed ours: for $100 \le n \le 5120$, $d_n(1/2)^{-2}$ differs by at most 5.1% from the least-squares fit $21.99 \log n$ of [5, §8]. This agreement also provides an independent check of the computation described in Subsection 4.1.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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